INSY7315

Work Integrated Learning

Task 2 AI Usage Report

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Generative AI was used during this phase of the project. This write-up explains where it was used, what it produced, and how that output was checked. We used Claude, mainly the Opus and Sonnet models, through Claude Code, a command-line tool with direct access to the project files, rather than through the web application.

1 WHERE AI WAS USED

Seed Data. The data to seed the database with during testing was AI-generated and further refined at subsequent data model changes.

Language Translations. The translations to isiXhosa and Afrikaans were done using machine translation using AI models. The Afrikaans translations were checked, but the isiXhosa translations were not able to be reviewied as we did not have a native speaker on our team.

Debugging. We used AI to trace the root causes of several subtle bugs, where it read the relevant code and data together and proposed explanations that we then confirmed.

Examples:

- **Release after the deadline.** A claim released after the offer's collection deadline reopened the offer and notified recipients about food that could no longer be collected. The release path now checks the deadline before reopening.
- **Dispatched driver left on a cancelled claim.** Releasing or expiring a claim did not cancel its collection, so a driver would still be sent to the donor. Release now cancels the active collection in the same transaction.
- **Notifications inside the claim transaction.** Writing notification rows inside the claim transaction kept the offer locked, and one failed write aborted a claim that had already won. Notifications are now sent after the claim commits.
- **Stuck failed requests.** A failed fetch was kept by the data client, so every later write failed the same way. The failure is no longer kept.
- **Remainder offers.** Creating the remainder offer inside the handover transaction aborted the whole handover on a duplicate reference. It is now created after the commit, and a repeated confirmation creates it if it is missing.

Tests AI was used to generate and further refine several test cases in finnicky areas such as concurrency and race conditions involving Claims, collections, handovers, webhooks, PDFs, and certificates. In addition, several infrastructure-dependent integration tests and time-dependent test cases were generated and/or reviewed and revised using AI.

These test cases were carefully reviewed and adjusted when needed before inclusion in the codebase.

README. We used AI to format and check the README for gaps.